

RAHUL P

Senior Python Developer | 6+ Years | Network Automation | GenAI | Chrome Extensions

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PROFESSIONAL SUMMARY

Senior Python Developer with 6+ years of experience shipping production backend systems, REST APIs, multi-vendor network automation platforms, and Chrome browser extensions for enterprise client Juniper Networks (HPE). Recently delivered a Generative AI Configuration Impact Analyzer powered by Large Language Models that predicts the downstream impact of network configuration changes before rollout, helping operators catch risky changes pre-deployment. Architected and shipped two internal Chrome extensions adopted by the full client developer team and recognized by client leadership. Strong expertise in Python, Flask, Django, FastAPI, PostgreSQL, MySQL, Docker, Kafka, and Linux, with deep specialization in SONiC, NX-OS, and Junos automation using NETCONF, Netmiko, and Ansible, plus Zero Touch Provisioning (ZTP) and IPv6. Fluent with modern AI-assisted development (GitHub Copilot, Cursor, Kiro, Claude Code, Trae), SonarQube static analysis, and AWS EC2 self-hosting. Active open-source contributor with 20+ public Python and Go projects on GitHub.

TECHNICAL SKILLS

Programming Languages: Python, Go, SQL, JavaScript, HTML, CSS, Bash, YAML

Web Frameworks: Flask, Django, FastAPI, Django REST Framework, Jinja2, SQLAlchemy

APIs and Protocols: REST APIs, GraphQL, JSON, XML, WebSockets

Databases: PostgreSQL, MySQL, SQLite, Redis

Network Automation: SONiC, NX-OS, Junos, Cisco, Apstra, Device Profiles, Interface Maps, Logical Devices, Zero Touch Provisioning (ZTP), IPv6, NETCONF, Ansible

Generative AI and LLMs: LLM integration (OpenAI, Anthropic Claude), prompt engineering, Retrieval-Augmented Generation (RAG), vector embeddings, AI-driven configuration impact analysis

AI Development Tooling: GitHub Copilot, Cursor, Kiro, Trae, Claude Code

Browser Extensions: Chrome Extension, background service workers, content scripts, cookie and session automation, chrome.storage, chrome.cookies, chrome.tabs

Web Technologies: HTML5, CSS3, JavaScript, JSON, WebSockets, Bootstrap

DevOps and CI/CD: Docker, Docker Compose, Kubernetes (basics), Jenkins, GitHub Actions, GitLab CI, Bitbucket Pipelines

Code Quality and Security: SonarQube static code analysis, code coverage, linting, secure coding practices, Metasploit / msfvenom (payload generation for security research)

Cloud Hosting: AWS EC2 (Linux VM provisioning, Nginx, systemd services, SSL via Let's Encrypt), S3, IAM, security groups, self-hosted web application deployment

Messaging and Queues: Apache Kafka

Version Control: Git, GitHub, GitLab, Bitbucket

Testing: pytest, unittest, mock, Ptest

Cloud and Platforms: AWS (EC2, S3, IAM basics), Linux (Ubuntu, Kali), Windows, WSL2

Methodologies: Agile, Scrum, Kanban, Code Reviews

PROFESSIONAL EXPERIENCE

Senior Software Developer | Calsoft (Client: Juniper Networks / HPE), Bangalore

June 2023 - Present

Stack: Python, Network Automation, SONiC, NX-OS, Junos, ZTP, IPv6, Docker, Generative AI, Apstra, Chrome Extensions, Kanban

- Delivered a Generative AI Configuration Impact Analyzer powered by Large Language Models that predicts the downstream impact of user-applied device configurations before rollout, helping operators catch risky changes pre-deployment.
- Architected and shipped a Chrome extension that monitors Jenkins build statuses across multiple servers in a single unified dashboard; initially built for personal productivity, later adopted by the entire client developer team and recognized by client leadership for measurable productivity gains.
- Architected and shipped a Chrome extension that automates internal portal logins via cookie-based session injection; rolled out from personal use to the full client developer team and praised by the client for eliminating repetitive authentication and saving daily engineering time.
- Designed and developed Device Profiles, Interface Maps, and Logical Devices that power scalable, automated network configuration across thousands of multi-vendor devices.
- Built configuration validation tooling for SONiC, NX-OS, and Junos platforms, ensuring cross-platform accuracy and reducing post-deployment configuration errors.
- Automated device onboarding through Zero Touch Provisioning (ZTP), reducing manual setup effort and accelerating deployment turnaround.
- Developed and integrated Impact APIs to streamline data flow across blueprint, topology, and configuration layers.
- Implemented Ptest frameworks that validate inter-port constraints and enforce compliance with network design standards.
- Diagnosed and resolved complex issues in network blueprints, topologies, and configurations, improving overall system stability.
- Ran SonarQube static analysis on project repositories to detect bugs, code smells, and security vulnerabilities, improving overall code health and maintainability.
- Accelerated feature delivery and code quality using AI-assisted development tools (GitHub Copilot, Cursor, Kiro, Claude Code, Trae) for prototyping, refactoring, and review.

Senior Software Engineer | Wipro (Client: Blackstone), Bangalore

June 2021 - December 2022

Stack: Python, Flask, REST APIs, MySQL, Docker, Bitbucket, Agile, Scrum

- Designed and built custom Flask applications and RESTful APIs aligned to client business requirements, delivered through two-week Agile sprint cycles.
- Collaborated with cross-functional teams on functional and technical analysis, system design, and implementation.
- Authored unit tests using pytest and mocking frameworks, raising code coverage and catching regressions early.
- Contributed to sprint planning, code reviews, refactoring, and deployment to development and testing environments.

Software Engineer | Accion Labs, Bangalore

November 2019 - May 2021

Stack: Python, Django, PostgreSQL, MySQL, GitLab, Docker, Agile

- Built a Health Insurance web application using Django, including custom template tags, filters, and decorators tailored to business workflows.
- Integrated PostgreSQL and MySQL for data persistence and implemented unit tests with the unittest and mock frameworks.
- Contributed to feature delivery, code reviews, and release activities across the web development team.

PROJECTS

- **Jenkins All-in-One Monitor (Chrome Extension)** - Originally built for personal productivity; adopted by the client team and rolled out to all developers. Aggregates build statuses from multiple Jenkins servers into a single dashboard with real-time notifications. Recognized by the client for measurable productivity gains.
- **Cookie-Based Auto-Login (Chrome Extension)** - Originally built for personal use; adopted by the client team and rolled out to all developers. Automates login to internal portals by injecting and managing session cookies, eliminating repetitive manual authentication. Praised by the client for saving daily engineering time.
- **GenAI Network Configuration Impact Analyzer (Python, LLMs)** - Uses Large Language Models to analyze proposed device configuration changes and predict their downstream impact prior to production rollout.
- **netmon - Network Monitor and Firewall (Go, Web UI)** - Single-binary, browser-based network monitor and firewall with per-process traffic inspection and live connection tracking. Self-hosted, zero dependencies.
- **Self-Hosted Web Applications on AWS EC2 (Linux, Nginx)** - Provisioned and hardened AWS EC2 instances to host personal Python and Go web applications, including reverse proxy with Nginx, systemd-managed services, and SSL certificates.
- **msfauto - Automated msfvenom Payload Generator (Python)** - Command-line tool that automates Metasploit msfvenom payload generation for authorized penetration testing and security research workflows.
- **GmailCleaner (Go)** - Self-hosted Gmail cleaning tool with AI-based email categorization, bulk email management, and multi-account support.
- **Dupcleaner (Go)** - Fast duplicate file finder using multi-algorithm detection (hash and perceptual) with a modern web user interface.
- **Naukri-Updater (Python)** - Automation tool that updates resumes on job portals with a single click, improving profile visibility for job seekers.
- **Real Mock CSV Generator (Python)** - Generates realistic mock datasets into CSV files within seconds for testing, prototyping, and load simulation.
- **Image Sorter / CVLib Image Sorter (Python, OpenCV)** - Sorts images by detected faces using OpenCV and cvlib facial detection pipelines.

EDUCATION

Bachelor of Technology (B.Tech) in Engineering - Sri Venkateshwara Institute of Technology, Anantapur, Andhra Pradesh

Higher Secondary (Intermediate) - Narayana Junior College, Anantapur, Andhra Pradesh

Secondary School Certificate (SSC) - C.V.R High School, Anantapur, Andhra Pradesh